Bees 4: honey bees and bumblebees on the same flowers

Simon Frost

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```
ENV["GKSwstype"] = "100"
using MatrixCommunityModels
using Plots
```

The previous three vignettes built a honey bee colony with *Varroa* and deformed wing virus, and a bumblebee population. Neither refers to the other. This vignette joins them through two kinds of pool: what the flowers offer, and what infected foragers leave on them. Because the link is a pool and not a line of code inside either species, it can be swapped: four different ideas of competition are four different couplings of the same two models.

Demand and supply are pools

Each species reports how much nectar and pollen its foragers would collect today (the pools `hb_nectar`, `hb_pollen`, `bb_nectar`, `bb_pollen`) and receives the fraction of that demand the flowers can meet. `floral_competition` supplies the rule; the supply itself is a function of the day of the year, in grams per square kilometre per day. The default, up to 3 kg of sugar and 3 kg of pollen a day, is of the order measured on intensively farmed land; flower-rich habitat offers 3 to 60 times as much (Hicks et al. 2016; Baude et al. 2016).

```
function landscape(mode; hives = 1, years = 6, virus = false, treated =
 ↪  true, kw...)
    forage = floral_competition(mode; hives = hives, kw...)
    flowers = virus ? flower_transmission(hives = hives) : nothing
    treatment = treated ? (day -> 270 <= day < 310 ? 0.115 : 0.0) : (day ->
 ↪  0.0)
    c = community(honeybee(virus = virus, forage = forage, flowers =
 ↪  flowers),
                  varroa(virus = virus, treatment = treatment),
                  bumblebee(virus = virus, forage = forage, flowers =
                   ↪  flowers), forage.species...)
    n0 = honeybee_state(c; mites = virus ? 100 : 0, viral = virus ? 0.1 :
 ↪  0.0, queens = 100)
    return c, project(c, n0, 1:(365 * years))
end

c, alone = landscape(:scramble; hives = 0)
species_names(c), pool_names(c)
```

```
([:honeybee, :varroa, :bumblebee], [:adult_bees, :brood, :capping,
 ↪  :emerging, :drone_capping, :drone_emerging, :hb_nectar, :hb_pollen,
 ↪  :bb_nectar, :bb_pollen, :phoretic])
```

`hives` is the number of honey bee colonies per square kilometre, each like the one modelled. National averages are 2.5 to 3.6 per square kilometre, and effects on wild bees have been measured at 4 to 14 (Henry & Rodet 2018). One colony takes far more than the whole bumblebee population:

```
day = 365 + 190
totals = pool_totals(c, alone.u[day + 1, :], day)
(honey_bee_colony = (nectar = round(totals.hb_nectar), pollen =
 ↪  round(totals.hb_pollen)),
 bumblebees = (nectar = round(totals.bb_nectar), pollen =
 ↪  round(totals.bb_pollen)))
```

```
(honey_bee_colony = (nectar = 0.0, pollen = 0.0), bumblebees = (nectar =
 ↪  49.0, pollen = 17.0))
```

(With `hives = 0` the colony's demand is reported as zero; multiply by one for a single colony.)

```
_, one = landscape(:scramble; hives = 1)
totals_one = pool_totals(landscape(:scramble; hives = 1)[1], one.u[day + 1,
 ↪  :], day)
round(totals_one.hb_nectar / totals_one.bb_nectar; digits = 1)
```

18.8

That ratio is why competition between the two is so asymmetric: bumblebees cannot deplete a landscape that feeds honey bees, but honey bees can deplete one that feeds bumblebees.

Four ways to compete

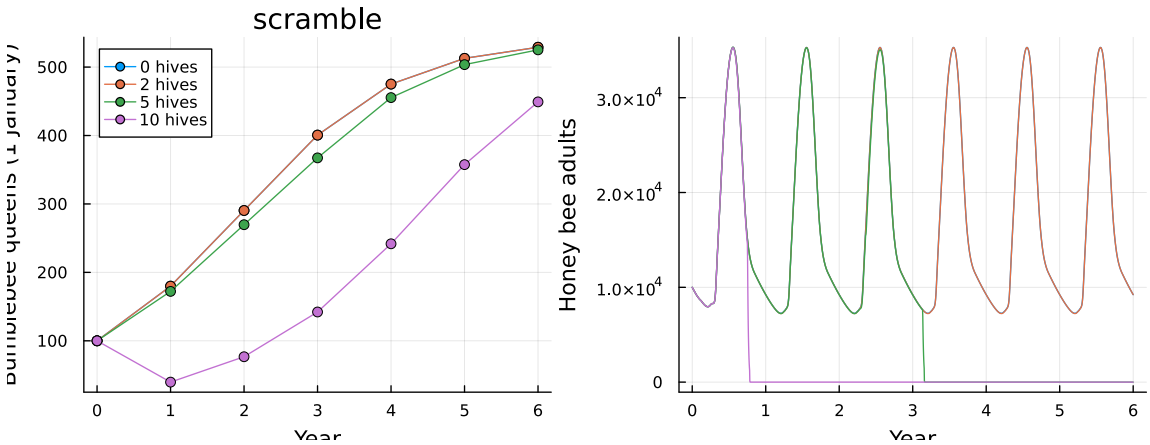
```
queens(sol) = stage_total(sol, :bumblebee, :hibernating)[365 .* (0:6) .+ 1]
workers(sol) = stage_total(sol, :honeybee, [:hive, :forager])

function compare(mode, densities; kw...)
    runs = [landscape(mode; hives = h, kw...)[2] for h in densities]
    p1 = plot(0:6, reduce(hcat, queens.(runs)); label = permutedims(["$h
 ↪  hives" for h in densities]), marker = :circle,
              xlabel = "Year", ylabel = "Bumblebee queens (1 January)",
 ↪  title = String(mode), legend = :topleft)
    p2 = plot(runs[1].t ./ 365, reduce(hcat, workers.(runs)); label = "",
 ↪  xlabel = "Year", ylabel = "Honey bee adults")
    plot(p1, p2; layout = (1, 2), size = (850, 320))
end
densities = [0, 2, 5, 10]
```

```
4-element Vector{Int64}:
  0
  2
  5
 10
```

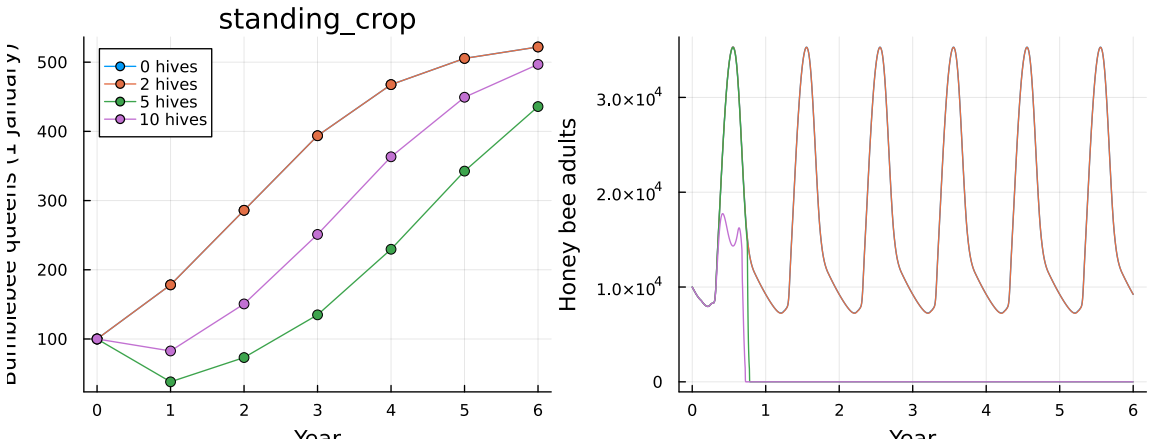
Scramble. Everyone gets the same fraction of what they ask for: `min(1, supply / total demand)`.

```
compare(:scramble, densities)
```



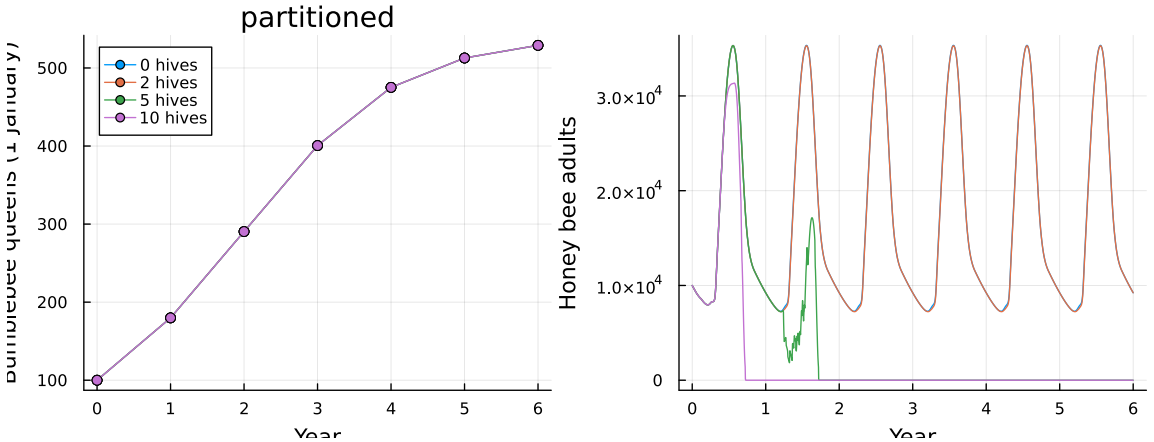
Standing crop. The supply is a stock that flowers refill each day and from which half of what is not taken carries over. Heavy use today leaves less tomorrow. This adds a `flowers` species to the community.

```
compare(:standing_crop, densities)
```



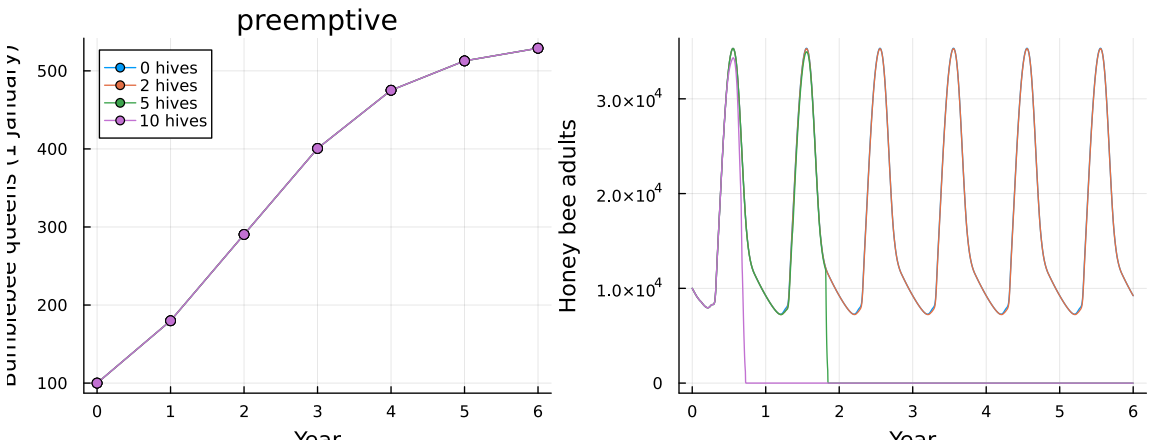
Partitioned. Half of the supply is open to both; the rest is split into flowers only honey bees use and flowers only bumblebees use.

```
compare(:partitioned, densities)
```

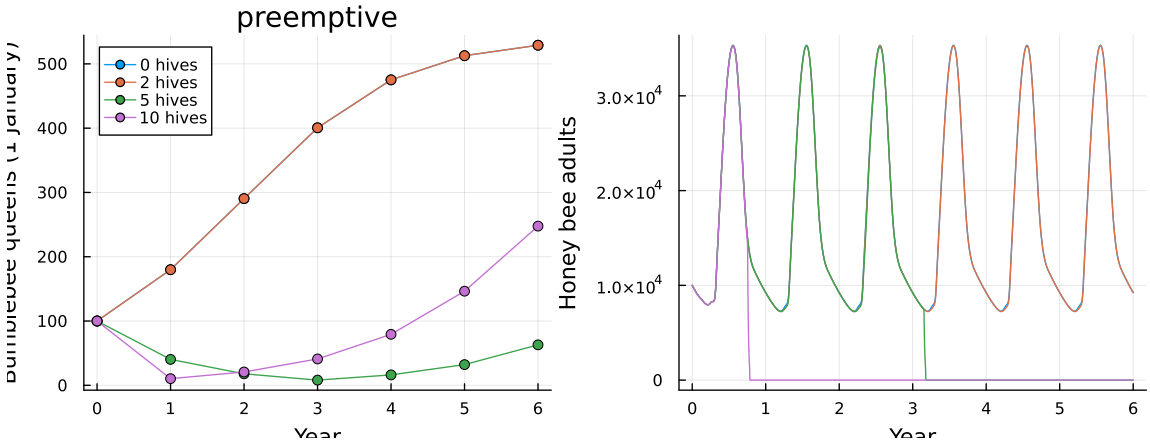


Pre-emptive. Honey bees, which recruit nestmates to rich patches, take what they want first; bumblebees get what is left, plus a refuge of deep flowers that honey bees cannot use. With a refuge of 20%, and with none:

```
compare(:preemptive, densities)
```




```
compare(:preemptive, densities; refuge = 0.0)
```



The four couplings tell different stories from the same two species:

```
final(mode; kw...) = [round(queens(landscape(mode; hives = h,
 ↪  kw...)[2])[end]; digits = 0) for h in densities]
(hives = densities, scramble = final(:scramble), standing_crop =
 ↪  final(:standing_crop),
 partitioned = final(:partitioned), preemptive = final(:preemptive),
 ↪  preemptive_no_refuge = final(:preemptive; refuge = 0.0))
```

```
(hives = [0, 2, 5, 10], scramble = [529.0, 529.0, 525.0, 449.0],
 ↪  standing_crop = [522.0, 522.0, 436.0, 497.0], partitioned = [529.0,
 ↪  529.0, 529.0, 529.0], preemptive = [529.0, 529.0, 529.0, 529.0],
 ↪  preemptive_no_refuge = [529.0, 529.0, 63.0, 248.0])
```

Three things stand out. At two hives per square kilometre nothing happens under any rule: the flowers offer more than everyone asks for. Where bumblebees have any flowers of their own, a partition or a refuge, honey bees do them no harm at any density, because bumblebee demand is

so small that a small exclusive share covers it. And where everything is shared and the hives ask for more than there is, the hives suffer first: identical colonies all run short together and die within a few years, after which the bumblebees recover. That is why ten hives can leave more bumblebees after six years than five do. Only where honey bees take first and there is no refuge are bumblebees badly set back.

These are consequences of the couplings, not findings about real landscapes. Reviews find negative effects on wild bees in about half of studies (Mallinger et al. 2017). Two results here do not match the field work. Near apiaries at more than 14 colonies per square kilometre, honey bees and wild bees lost foraging success together, by about half each (Henry & Rodet 2018); hives did not collapse first, and beekeepers feed or move colonies that are short. And bumblebee colonies beside apiaries gain less weight and rear fewer and smaller queens even in ordinary vegetation, where the model with any refuge predicts no effect. Worker size, through which much of that effect runs, is not modelled.

When the flowers stop

Bumblebees keep no stores, so *when* forage is available matters more to them than to honey bees. Ending the season's forage two months early, with one hive:

```
supply(finish) = (nectar = floral_supply(3000.0; finish = finish), pollen =
 ↪  floral_supply(3000.0; finish = finish))
_, usual = landscape(:scramble; hives = 1, supply(280)...)
_, short = landscape(:scramble; hives = 1, supply(220)...)
(bumblebee_queens = (usual = round(queens(usual)[end]), short_season =
 ↪  round(queens(short)[end])),
 honey_bee_peak = (usual = round(maximum(workers(usual)[(365 * 5):end])),
 ↪  short_season = round(maximum(workers(short)[(365 * 5):end]))))
```

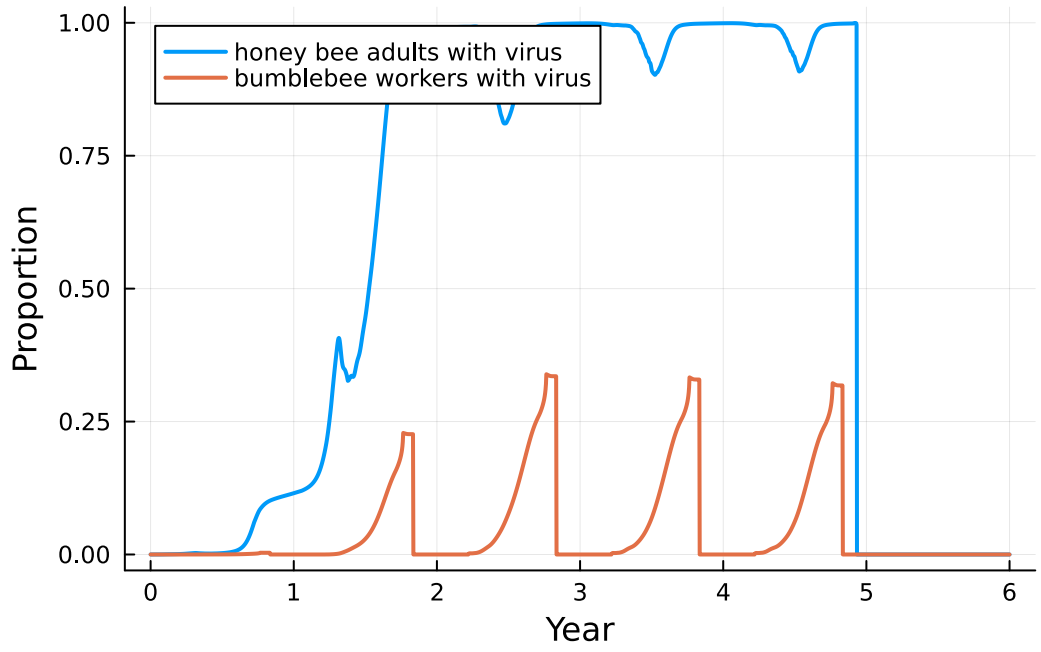
```
(bumblebee_queens = (usual = 529.0, short_season = 469.0), honey_bee_peak =
 ↪  (usual = 35295.0, short_season = 35045.0))
```

The hive, which lives on its stores, is unaffected. The bumblebee population ends lower: colonies founded by late-emerging queens are still rearing brood when the forage stops.

Virus on flowers

The second kind of pool carries the virus. Honey bee foragers infected as pupae report their flower visits to `infectious_visits`; an infected forager leaves virus on about ten flowers a day, the fraction of flowers contaminated is `1 - exp(-contaminating visits / flowers)`, so more flowers dilute it; and a bumblebee worker picks the virus up with a daily probability about equal to that fraction. Bumblebees have no mites, do not (by default) pass the virus on, and are not harmed by it unless `infected_hazard` is raised.

```
cv, spill = landscape(:scramble; hives = 1, virus = true, treated = false)
bb_prev = stage_total(spill, :bumblebee, :infected_worker) ./
          max.(stage_total(spill, :bumblebee, [:worker, :infected_worker]),
           ↪  1e-9)
hb_prev = stage_total(spill, :honeybee, [:hive_infected, :forager_infected,
 ↪  :hive_carrier, :forager_carrier]) ./ max.(workers(spill), 1.0)
plot(spill.t ./ 365, [hb_prev bb_prev]; label = ["honey bee adults with
 ↪  virus" "bumblebee workers with virus"],
     xlabel = "Year", ylabel = "Proportion", lw = 2)
```




```
summer = 365 .* (0:5) .+ 200
[(year = y, honey_bees = round(hb_prev[d]; digits = 2), bumblebees =
 ↪  round(bb_prev[d]; digits = 2)) for (y, d) in enumerate(summer)]
```

```
6-element Vector{@NamedTuple{year::Int64, honey_bees::Float64,
 ↪  bumblebees::Float64}}:
 (year = 1, honey_bees = 0.0, bumblebees = 0.0)
 (year = 2, honey_bees = 0.54, bumblebees = 0.05)
 (year = 3, honey_bees = 0.86, bumblebees = 0.11)
 (year = 4, honey_bees = 0.91, bumblebees = 0.11)
 (year = 5, honey_bees = 0.91, bumblebees = 0.1)
 (year = 6, honey_bees = 0.0, bumblebees = 0.0)
```

The untreated hive dies in its fifth winter, and the virus goes with it. While the hive lives, bumblebee prevalence follows it, with no route of its own. When the hive is treated for mites, the virus all but disappears from both:

```
_, treated = landscape(:scramble; hives = 1, virus = true, treated = true)
maximum(stage_total(treated, :bumblebee, :infected_worker) ./
        max.(stage_total(treated, :bumblebee, [:worker, :infected_worker]),
         ↪  1e-9))
```

```
0.003258155371357223
```

In field surveys DWV is found in 36 to 45% of honey bees and 3 to 11% of bumblebees at the same sites, in 16% of bumblebees where there is an apiary against 5.5% where there is none within a kilometre, and is far more prevalent in both where *Varroa* is present (Fürst et al. 2014; McMahon et al. 2015; Manley et al. 2019; Alger et al. 2019). The transmission rates here were chosen to give about a tenth of bumblebee workers in midsummer; they are not measured.

Three things the model leaves out bear on these numbers. A bumblebee that has swallowed the virus is counted as infected for life, but in experiments the virus mostly stays in the gut without multiplying, and many bees lose it within days (Gusachenko et al. 2020; Streicher et al. 2024; Burnham et al. 2021), so the model's prevalence is best compared with the share of bees testing positive, not with infections. Treated hives lose the virus entirely here, because mites are the only route between honey bees; real treated colonies still carry it, and a few percent of nearby bumblebees would be expected to. And virus is taken to last a day on a flower; how long it lasts is not known.

What is provisional

The supply and the conversion from honey to sugar removed now rest on measurements (Hicks et al. 2016; Baude et al. 2016), though for one kind of landscape and with a flat season, where farmland supply has a gap in June and falls from mid-August (Timberlake et al. 2019). The overlap and refuge fractions, the carry-over of the standing crop and the transmission rates are still choices: no measurements were found for them. *Bombus terrestris* has a short tongue, so a refuge of deep flowers fits it poorly, and the no-refuge case may be the fairer one for this species. Of the papers cited here, Hicks, Baude, Henry & Rodet, Mallinger, McMahon, Manley, Alger and Streicher were read; the others are known from abstracts, summaries or released data. What the vignette shows is that the alternatives can be stated, swapped and compared without touching either species' model.

References

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Royal Society B 291, 20232293. Timberlake, T. P., Vaughan, I. P. & Memmott, J. (2019). *Journal of Applied Ecology* 56, 1585-1596.